

SQL COMPLETE GUIDE — PART 2

ADVANCED SQL → OBJECTS → TRANSACTIONS → PERFORMANCE → DESIGN → SECURITY →
WAREHOUSING → AIML

21. Views

View — Definition: A named query that can be queried like a table; a normal view generally stores the query definition rather than a separate copy of data.

Materialized View — Definition: A query result physically stored and refreshed where supported.

Learn updatable views, security through views, WITH CHECK OPTION where supported and materialized-view refresh.

```
CREATE VIEW Active_Customers AS
SELECT Customer_ID, Name, Email
FROM Customers WHERE Status='Active';
```

22. Stored Procedures

Stored Procedure — Definition: A named database program containing SQL statements and procedural logic.

Parameter — Definition: A value passed into or returned from a procedure/function.

Learn parameters, variables, IF/CASE, loops, error handling and transaction behavior. Syntax varies by DBMS.

23. User-Defined Functions

UDF — Definition: A custom function created by a developer that returns a value; some DBMSs also support table-valued functions.

Learn scalar functions, table-valued functions where supported, parameters and DBMS-specific restrictions.

24. Triggers

Trigger — Definition: A database program automatically executed when a supported event such as INSERT, UPDATE or DELETE occurs.

BEFORE Trigger — Definition: Runs before the event where supported.

AFTER Trigger — Definition: Runs after the event.

INSTEAD OF Trigger — Definition: Replaces an operation, commonly used with some views.

Uses: audit logging, history and specialized validation. Caution: triggers can hide business logic and complicate debugging.

25. Temporary Tables

Temporary Table — Definition: An intermediate table usually scoped to a session or transaction depending on DBMS.

Compare CTE = one statement; temporary table = reusable intermediate object in its scope; view = persistent query interface.

26. Window Functions

Window Function — Definition: Calculates across related rows while keeping individual rows in the result.

OVER() — Definition: Defines the window.

PARTITION BY — Definition: Divides rows into independent groups without collapsing them.

ROW_NUMBER — Definition: Unique sequential number within a window.

RANK — Definition: Ties share rank and gaps appear after ties.

DENSE_RANK — Definition: Ties share rank without gaps.

LAG — Definition: Reads a previous row's value.

LEAD — Definition: Reads a following row's value.

FIRST_VALUE/LAST_VALUE — Definition: Returns first/last value according to the window.

```
SELECT Name, Department, Salary,  
       ROW_NUMBER() OVER(PARTITION BY Department ORDER BY Salary DESC) rn,  
       RANK() OVER(PARTITION BY Department ORDER BY Salary DESC) rnk,  
       DENSE_RANK() OVER(PARTITION BY Department ORDER BY Salary DESC) drnk  
FROM Employees;
```

```
SELECT Sale_Date, Sales,  
       SUM(Sales) OVER(ORDER BY Sale_Date) Running_Total,  
       LAG(Sales) OVER(ORDER BY Sale_Date) Previous_Sales  
FROM Daily_Sales;
```

27. Window Frames

Window Frame — Definition: The subset of rows considered for the current row, commonly defined with ROWS or RANGE.

Learn ROWS vs RANGE, preceding/following, current row and unbounded frames.

```
SUM(Sales) OVER(  
  ORDER BY Sale_Date  
  ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW  
) AS Running_Total
```

28. Transactions and TCL

Transaction — Definition: A logical unit of database work that should succeed or fail as a unit.

COMMIT — Definition: Makes transaction changes permanent.

ROLLBACK — Definition: Undoes uncommitted changes.

SAVEPOINT — Definition: Creates an intermediate rollback point.

```
START TRANSACTION;  
UPDATE Accounts SET Balance=Balance-1000 WHERE Account_ID=1;  
UPDATE Accounts SET Balance=Balance+1000 WHERE Account_ID=2;  
COMMIT;
```

29. ACID

Property	Definition	Example
Atomicity	All operations succeed or none do	Debit + credit both succeed
Consistency	Rules remain valid	Constraints remain valid
Isolation	Concurrent work is controlled	Transfers don't interfere incorrectly
Durability	Committed changes survive failure	Committed transfer remains

30. Isolation, Locks and Concurrency

Isolation Level — Definition: Controls how much one transaction is isolated from concurrent transactions.

Level	Typical behavior
READ UNCOMMITTED	May allow dirty reads
READ COMMITTED	Prevents dirty reads; non-repeatable reads may occur
REPEATABLE READ	Stronger repeatability; exact behavior varies
SERIALIZABLE	Strongest standard isolation; lower concurrency

Dirty Read — Definition: Reading uncommitted changes from another transaction.

Non-repeatable Read — Definition: Reading the same row twice and getting different committed values.

Phantom Read — Definition: Repeating a range query and seeing new/removed rows.

Lock — Definition: A concurrency-control mechanism coordinating access to data.

Deadlock — Definition: Transactions wait for each other's resources and cannot proceed until resolved.

31. Indexes

Index — Definition: A data structure that can make suitable data lookups faster.

Single-column index — Definition: An index on one column.

Composite index — Definition: An index on multiple columns; column order matters.

Unique index — Definition: An index that enforces uniqueness when configured to do so.

Clustered index — Definition: An index organization where table storage is organized around an index key in systems supporting the concept.

Non-clustered index — Definition: A separate index structure pointing to underlying rows.

```
CREATE INDEX idx_orders_customer ON Orders(Customer_ID);
CREATE INDEX idx_orders_customer_date
ON Orders(Customer_ID, Order_Date);
```

Indexes speed suitable reads but consume storage and add write maintenance.

32. EXPLAIN and Query Plans

EXPLAIN — Definition: Shows the planned execution strategy; some DBMSs also provide actual execution plans.

Query Plan — Definition: The strategy chosen by the optimizer to execute a query.

Cost — Definition: An optimizer estimate used to compare plans; not necessarily elapsed time.

```
EXPLAIN
SELECT * FROM Orders
WHERE Customer_ID=1001;
```

Check scans, index usage, join strategy, row estimates, sorting, aggregation and expensive operators.

33. Query Optimization

- Select only required columns where practical.
- Use selective filters and appropriate indexes.
- Check JOIN conditions to avoid accidental Cartesian products.
- Avoid unnecessary DISTINCT and sorting.
- Avoid implicit type conversions.
- Compare execution plans before and after changes.
- Test with realistic data volume.
- Use appropriate pagination for large result sets.

34. Normalization

Normalization — Definition: Organizing relational data to reduce unnecessary duplication and update anomalies.

Form	Definition
1NF	Atomic values; no repeating groups
2NF	1NF + no partial dependency on part of a composite key
3NF	2NF + no transitive dependency
BCNF	Every determinant is a candidate key
4NF	Addresses certain multi-valued dependencies
5NF	Addresses certain join dependencies; advanced

Denormalization — Definition: Intentionally adding controlled redundancy for performance or analytics simplicity.

35. Relationships and Referential Integrity

One-to-One — Definition: One row in A relates to at most one row in B.

One-to-Many — Definition: One row in A relates to many rows in B.

Many-to-Many — Definition: Many rows in A relate to many rows in B, usually through a junction table.

Referential Integrity — Definition: Ensures foreign-key values reference valid parent keys according to constraints.

Foreign-key actions can include CASCADE, SET NULL, SET DEFAULT or RESTRICT/NO ACTION where supported.

36. Advanced Query Patterns

Conditional Aggregation — Definition: Using CASE inside aggregates to calculate multiple conditional metrics.

Anti Join — Definition: Finds rows without a match, commonly LEFT JOIN + IS NULL or NOT EXISTS.

Semi Join — Definition: Finds rows with a match, commonly EXISTS.

Top-N per Group — Definition: Finds top N rows inside each group using ranking window functions.

Running Total — Definition: Cumulative total over an ordered window.

Moving Average — Definition: Average over a sliding window.

Gaps and Islands — Definition: Techniques for consecutive runs, missing periods and groups.

```
WITH Ranked AS(
  SELECT *,
    ROW_NUMBER() OVER(PARTITION BY Department ORDER BY Salary DESC) rn
  FROM Employees
)
SELECT * FROM Ranked WHERE rn<=3;
```

37. Recursive CTE and Hierarchies

Hierarchy — Definition: Parent-child data such as employee-manager or category-subcategory relationships.

```
WITH RECURSIVE Org AS(
  SELECT Employee_ID, Name, Manager_ID, 0 AS Level
  FROM Employees WHERE Manager_ID IS NULL
  UNION ALL
  SELECT e.Employee_ID, e.Name, e.Manager_ID, o.Level+1
  FROM Employees e JOIN Org o ON e.Manager_ID=o.Employee_ID
)
SELECT * FROM Org;
```

38. Partitioning

Table Partitioning — Definition: Splits a large logical table into smaller physical partitions while keeping one logical table interface.

Type	Concept
Range	Ranges such as dates
List	Discrete values such as country
Hash	Hash of a key
Composite	Combination of methods

Scenario: A telecom alarm table with billions of rows may be partitioned by date/month for manageability and partition pruning.

39. JSON and Advanced Data Types

JSON — Definition: Semi-structured data type/functions supported by many databases.

Array — Definition: A column containing multiple values in systems supporting arrays.

UUID — Definition: A globally unique identifier often used instead of sequential numeric IDs.

ENUM — Definition: A restricted predefined set of values where supported.

Spatial Data — Definition: Geographic/geometry data used for location applications where supported.

For AIML/data engineering, learn JSON extraction, nested fields, casting and validation.

40. PIVOT / UNPIVOT

PIVOT — Definition: Transforms row categories into columns for cross-tab reporting where supported.

UNPIVOT — Definition: Transforms columns into rows where supported.

```
SELECT Department,  
       SUM(CASE WHEN Status='Active' THEN 1 ELSE 0 END) Active_Count,  
       SUM(CASE WHEN Status='Inactive' THEN 1 ELSE 0 END) Inactive_Count  
FROM Employees  
GROUP BY Department;
```

41. SQL Security

GRANT — Definition: Gives permissions to a user or role.

REVOKE — Definition: Removes permissions.

Role — Definition: A named collection of permissions assigned to users.

Least Privilege — Definition: Give an account only the permissions it needs.

SQL Injection — Definition: A vulnerability where untrusted input changes intended SQL; prevent it with parameterized queries/prepared statements and least-privilege accounts.

```
GRANT SELECT ON Employees TO analyst_role;  
REVOKE SELECT ON Employees FROM analyst_role;
```

42. Data Warehousing

Fact Table — Definition: Stores measurable business events such as sales, transactions or alarms.

Dimension Table — Definition: Stores descriptive attributes such as customer, product, site or date.

Star Schema — Definition: A fact table connected directly to dimension tables.

Snowflake Schema — Definition: A more normalized dimensional model.

ETL — Definition: Extract → Transform → Load.

ELT — Definition: Extract → Load → Transform.

43. Data Quality

Duplicate Detection — Definition: Finding repeated records based on a business key.

Missing Data Check — Definition: Finding required NULL or invalid values.

Orphan Detection — Definition: Finding child records whose parent does not exist.

Reconciliation — Definition: Comparing datasets to identify missing, extra or mismatched records.

```
SELECT Email,COUNT(*) AS cnt  
FROM Customers  
GROUP BY Email  
HAVING COUNT(*)>1;
```

44. AIML / Feature Engineering with SQL

Feature Engineering in SQL — Definition: Creating model-ready variables from database data using joins, filters, aggregations, dates, CASE and window functions.

```
SELECT Customer_ID,  
       COUNT(*) AS Order_Count,  
       SUM(Amount) AS Total_Spend,  
       AVG(Amount) AS Avg_Order_Value,  
       MAX(Order_Date) AS Last_Order_Date  
FROM Orders  
GROUP BY Customer_ID;
```

Important AIML topics: time windows, previous-period values, rolling averages, customer aggregates, missing-value handling, deduplication, labels and timestamp-based leakage checks.

45. Complete Scenario Bank

- E-commerce: top products, repeat customers, inactive customers, monthly revenue, customer lifetime spend.

- Telecom: latest device log, devices not updated in 24 hours, alarm count per site, critical alarms, zero-updating sites, daily trends.
- Banking: running balance, inactive accounts, transaction velocity, loan-without-account, branch metrics.
- HR: salary by department, top-N employees per department, manager hierarchy, salary bands.
- AIML: feature table, rolling averages, previous-month comparison, label creation, missing feature detection, leakage checks.

46. Complete Interview Questions

- 1. What are DQL, DML, DDL, DCL and TCL?
- 2. What is a schema?
- 3. DELETE vs TRUNCATE vs DROP?
- 4. WHERE vs HAVING?
- 5. INNER vs LEFT vs FULL JOIN?
- 6. What is a CROSS JOIN?
- 7. What is a SELF JOIN?
- 8. What are primary, foreign, candidate, composite and surrogate keys?
- 9. What are constraints?
- 10. What is a subquery?
- 11. What is a correlated subquery?
- 12. EXISTS vs IN?
- 13. ANY vs ALL?
- 14. What is a CTE?
- 15. CTE vs subquery vs temp table?
- 16. What is a recursive CTE?
- 17. What are SQL functions?
- 18. What is CASE?
- 19. What is a view?
- 20. View vs materialized view?
- 21. What is a stored procedure?
- 22. What is a UDF?
- 23. What is a trigger and when should you avoid it?
- 24. What are window functions?
- 25. ROW_NUMBER vs RANK vs DENSE_RANK?
- 26. LAG vs LEAD?
- 27. What is PARTITION BY?
- 28. What is a window frame?
- 29. What is a transaction?
- 30. Explain ACID.
- 31. What are isolation levels?
- 32. What is a dirty read?
- 33. What is a phantom read?
- 34. What is a deadlock?
- 35. What is an index?
- 36. What is a composite index?
- 37. Clustered vs non-clustered index?
- 38. What is EXPLAIN?
- 39. How do you optimize a slow query?
- 40. What is normalization?
- 41. Explain 1NF, 2NF, 3NF and BCNF.
- 42. What is denormalization?
- 43. What is referential integrity?
- 44. What is partitioning?
- 45. What are fact and dimension tables?

- 46. ETL vs ELT?
- 47. What is SQL injection and how do you prevent it?
- 48. How do you find duplicates?
- 49. How do you find orphan records?
- 50. How do you find the latest row per device?
- 51. How do you find top 3 records per group?
- 52. How do you calculate month-over-month growth?
- 53. How would you create an ML feature table using SQL?

47. Final Learning Order

Level	Topics
1	Database, schema, table, row, column, data types, keys
2	SELECT, aliases, DISTINCT, WHERE, operators, NULL, LIKE, IN, BETWEEN
3	ORDER BY, LIMIT/TOP/FETCH, CASE
4	INSERT, UPDATE, DELETE, MERGE
5	CREATE, ALTER, DROP, TRUNCATE, RENAME
6	Functions, dates, NULL handling
7	Aggregates, GROUP BY, HAVING
8	JOINS
9	Subqueries, EXISTS, ANY, ALL
10	CTEs and recursive CTEs
11	Views, materialized views, procedures, functions, triggers, temp tables
12	Window functions and frames
13	Transactions, ACID, isolation, locks, deadlocks
14	Keys, constraints, relationships, normalization
15	Indexes, EXPLAIN, optimization
16	JSON, PIVOT/UNPIVOT, partitioning, advanced query patterns
17	Security, roles, SQL injection
18	ETL/ELT, warehousing, data quality, AIML feature engineering
19	Scenario-based projects and interviews

48. Final SQL Mental Model

```

SCENARIO
↓
Identify tables + keys + relationships
↓
FROM / JOIN
↓
WHERE
↓
GROUP BY + aggregates
↓
HAVING
↓
CASE / functions
↓
Subquery / CTE / EXISTS if needed
↓
Window function if row-level analytics is needed
↓
ORDER BY / LIMIT

```

↓
Validate result
↓
EXPLAIN / optimize

Important: SQL syntax differs between MySQL, PostgreSQL, SQL Server and Oracle. Learn the concept first, then the dialect used in your project/interview.